

1. Overview of the Dataset:

The dataset contains information about patients and their health metrics aimed at predicting the presence of diabetes.

- **Number of rows:** 768
- **Number of columns:** 9
- **Goal:** Predict the presence of diabetes based on various health features.

2. Key Features:

The dataset includes the following key features:

- **Pregnancies:** Number of times a patient has been pregnant. This is significant as studies have shown that a higher number of pregnancies may increase the risk of developing diabetes.
- **Glucose:** Plasma glucose concentration after 2 hours in an oral glucose tolerance test. High glucose levels are associated with diabetes.
- **Blood Pressure:** Diastolic blood pressure (mm Hg). High blood pressure (hypertension) can be a risk factor for diabetes.
- **Skin Thickness:** Triceps skinfold thickness (mm). This indicates body fat levels, and higher values suggest obesity, which is linked to diabetes risk.
- **Insulin:** 2-hour serum insulin (μ U/ml). Issues with insulin production or function can lead to diabetes.
- **BMI (Body Mass Index):** Calculated as $\text{weight in kg} / (\text{height in m})^2$. High BMI, indicating overweight or obesity, increases the risk of diabetes.
- **Diabetes Pedigree Function:** A function that quantifies the likelihood of diabetes based on family history.
- **Age:** Age in years. Age is associated with increased risk of diabetes due to changes in glucose metabolism and insulin function.
- **Outcome:** This is the target variable, where 1 indicates the presence of diabetes, and 0 indicates no diabetes. 35% of patients in the dataset have diabetes (1).

3. Basic Statistics:

The basic statistics for each feature are as follows:

- **Pregnancies:** Average = 3.84, Range = [0, 17]
- **Glucose:** Average = 120.89, Range = [0, 199]
- **Blood Pressure:** Average = 69.11, Range = [0, 122]

- **BMI:** Average = 31.99, Range = [0, 67.1]
- **Age:** Average = 33.24, Range = [21, 81]

4. Data Cleaning:

Although the dataset contains no missing values, some features such as Glucose, Blood Pressure, Skin Thickness, and Insulin have zero values, which may indicate missing or erroneous data. These need further investigation.

5. Data Visualization:

Several visualizations were employed to explore the dataset:

- **Histograms, Box Plots, and Violin Plots** were used to analyze the distribution of key variables like Glucose, Blood Pressure, and BMI.
- **Count Plots and Pie Charts** were used to display the distribution of the Outcome variable (diabetes or no diabetes).

6. Analysis of Features with Outcome:

The relationship between each feature (e.g., Glucose, Blood Pressure, and BMI) and the outcome (diabetes or not) was analyzed using visualizations, helping to identify important factors influencing the likelihood of diabetes.

7. Correlation Analysis:

A correlation heatmap was created to analyze the relationships between features. The key correlations were:

- **Glucose** and **BMI** showed a strong correlation with the outcome.
- **Age** and **Pregnancies** also showed significant correlations.

8. Data Splitting:

The dataset was split into training (80%) and testing (20%) sets using `train_test_split` for model development.

9. Machine Learning Models:

Three machine learning models were trained on the dataset to predict diabetes:

- **AdaBoost Classifier:** Achieved an accuracy of 72.7%.
- **Logistic Regression:** Achieved an accuracy of 74.6%.
- **Random Forest Classifier:** Achieved an accuracy of 74%.

Each model's performance was measured using accuracy, recall, and F1 scores, and confusion matrices were used to visualize classification errors.

10. Prediction for High-Risk Individuals:

A Logistic Regression model was used to predict individuals who are at risk of developing diabetes based on their health metrics, even if they are not currently diagnosed with diabetes. The model flagged several individuals as high-risk based on their feature values.

Tools Used:

- **pandas, numpy:** For data manipulation and analysis.
- **matplotlib, seaborn, plotly:** For data visualization.
- **scikit-learn:** For machine learning model development (Random Forest, Logistic Regression, AdaBoost).
- **pickle:** To save the trained models for future use.

Conclusion:

This documentation gives a comprehensive overview of the steps involved in analyzing the diabetes dataset, from data cleaning to applying machine learning models for prediction.

Important Links

https://github.com/elbadry123/diabetes_disease.git